

Machine Learning Lifecycle: A Survey

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Abstract

The operationalization of machine learning (ML) introduces distinct engineering and lifecycle management challenges—such as extreme data dependence, silent model degradation (concept drift), and inherent non-determinism—which traditional software engineering workflows fail to adequately address. This systematic literature review provides a rigorous, comprehensive mapping of the ML lifecycle domain between 2015 and 2025 using the PRISMA protocol. Out of an initial pool of 12,450 articles, a highly specialized cohort of 22 primary studies was extracted, classified, and synthesized to map out contemporary Machine Learning Operations (MLOps) patterns, technical debt structures, governance models, and security vulnerabilities. To address the documented “production gap,” this paper formalizes the findings into a synthesized operational mapping and introduces a preliminary conceptual layout for an Adaptive Lifecycle Framework (ALF), juxtaposing it with legacy paradigms like CRISP-DM. Furthermore, we expand the scope to investigate domain-specific lifecycle complexities in healthcare systems and Large Language Model (LLM) pipelines, providing an essential evolutionary baseline for sustainable MLOps.

Keywords: machine learning; lifecycle management; MLOps; systematic review; technical debt; PRISMA

1. Introduction

The adoption of Machine Learning (ML) technology within industrial processes and computational science has rapidly evolved beyond experimental prototyping to become a core necessity for structural performance optimization [1]. Nonetheless, ML applications remain uniquely complex due to their structural divergence from deterministic software, characterized by intricate data dependencies, downstream model obsolescence, and systemic non-determinism [2,3]. Previous literature highlights that while advanced mathematical modeling provides exceptional prediction capabilities under idealized, static testing conditions, deploying and maintaining these architectures within dynamic production systems remains an arduous engineering hurdle, historically termed the “production gap” [4,5].

Many organizations attempt to mitigate these production challenges by relying on ad hoc trial-and-error procedures or by forcefully adapting legacy frameworks such as CRISP-DM (CRoss Industry Standard Process for Data Mining). However, these historical models are structurally unequipped to manage the continuous integration, continuous deployment (CI/CD), automated retraining, and regulatory governance demands of modern Machine Learning Operations (MLOps) [3,6]. This methodological heterogeneity introduces a significant risk: the lack of a standardized, unified lifecycle model that seamlessly



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merges technical automation infrastructure with robust governance controls compromises long-term system sustainability and introduces hidden technical debt [5,7].

To systematically investigate this landscape and provide actionable engineering clarity, this survey establishes and addresses four central Research Questions (RQs):

- RQ1: What are the foundational structural phases and operational components defining modern MLOps frameworks in the literature?
- RQ2: How does hidden technical debt manifest across the systemic non-code infrastructure of the ML lifecycle?
- RQ3: To what extent have ethical considerations, explainability (XAI), and security vulnerabilities been systematically incorporated into existing lifecycle designs?
- RQ4: How must traditional lifecycle paradigms evolve to accommodate highly specialized environments, such as medical pipelines and generative artificial intelligence (GenAI) architectures?

This systematic survey maps the empirical evolution of the ML lifecycle from 2015 to 2025 by critically analyzing 22 benchmark papers. Building upon the foundational research gaps identified by historical overviews [1,8], this work establishes an evolutionary baseline, provides a detailed quality-scored synthesis of the existing literature, and introduces a preliminary conceptual model for an Adaptive Lifecycle Framework designed to mitigate contemporary infrastructure debt.

2. Research Method and Search Strategy

2.1. Systematic Literature Review Protocol

This study implements a Systematic Literature Review (SLR) methodology meticulously aligned with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) protocol [8]. The research architecture relies on a multi-stage deterministic filtering process to identify, screen, and isolate high-impact, quality-verified primary academic contributions. The objective is to map out empirical shifts, structural taxonomies, and clear research deficits characterizing the ML lifecycle domain over the past decade.

2.2. Information Sources and Bibliographic Search Expression

To guarantee full domain coverage across computing, applied mathematics, and software engineering, comprehensive queries were executed across three major scientific databases: ScienceDirect (Elsevier), IEEE Xplore, and Google Scholar. The standardized Boolean search string was defined as:

“machine learning” AND “lifecycle”) OR (“MLOps”) OR (“ML pipeline management”)

The temporal parameters were strictly restricted to the past decade (2015–2025 inclusive) to isolate literature focusing on the modern transition from pure theoretical modeling to industrialized production engineering. The initial un-filtered database queries yielded a total of 12,450 candidate publications (ScienceDirect: 5120; IEEE Xplore: 4210; Google Scholar, post-filtering: 3120).

2.3. Inclusion and Exclusion Criteria

To maintain absolute methodological consistency while accurately representing the true multi-faceted nature of the field, eligibility criteria were refined to capture foundational frameworks and industry-validated empirical analyses:

- Inclusion Criteria (IC): (1) Original research papers published in peer-reviewed journals or seminal, peer-reviewed international conference proceedings (e.g., NeurIPS, MLSys, VLDB) that established domain definitions; (2) Studies introducing original operational frameworks, structural taxonomies, detailed case studies, or validated empirical lifecycles; and (3) Full-text manuscripts published entirely in English.

- Exclusion Criteria (EC): (1) Generalized white papers, commercial product brochures, or un-reviewed corporate marketing briefs; (2) Narrow algorithmic optimization papers focused entirely on mathematical hyperparameter tuning without a broader lifecycle, data pipeline, or MLOps infrastructure management context.

2.4. Study Selection and PRISMA Screening Flow

Applying the multi-stage PRISMA screening flow successfully reduced the initial pool of 12,450 records to a refined corpus of 22 benchmark studies. Screening proceeded in four stages: (i) Identification—12,450 records retrieved; (ii) Deduplication—1380 duplicates removed, leaving 11,070; (iii) Title/abstract screening—10,790 excluded, leaving 280 for full-text review; and (iv) Full-text eligibility assessment—258 excluded (214 not matching the IC/EC; 44 with quality scores below the threshold), leaving 22 included studies. Full details of each stage are provided in Sections 2.3 and 2.5.

2.5. Methodological Quality Assessment Framework

To eliminate the subjectivity typically associated with qualitative reviews, every selected study was subjected to a standardized Quality Assessment (QA) protocol grading five critical dimensions on a 3-point scale (0 = Insufficient, 1 = Partially Addressed, 2 = Fully Addressed): (QA1) Clarity of research aims; (QA2) Appropriateness of methodology; (QA3) Industrial/empirical validation; (QA4) Clarity of reported findings; and (QA5) Relevance to RQ1–RQ4. Studies scoring below 5/10 were excluded at the eligibility stage, and inter-rater disagreements were resolved by the third author.

Synthesized scoring revealed that modern peer-reviewed journal papers (e.g., [3,6,9]) demonstrated high overall scores due to detailed validation, while pioneering industrial technical reports (e.g., [10]) provided high pragmatic value despite lower formal academic methodology scoring.

3. Results and Literature Synthesis

3.1. Bibliometric and Taxonomic Distributions

Chronological mapping of the corpus clearly identifies a sharp publication inflection point beginning in 2020 [3,6]. This shift marks the transition of the academic and industrial community's focus away from localized model optimization toward end-to-end operational automation (MLOps).

3.1.1. Temporal Trends

Statistical evaluation of the corpus indicates that approximately 72.73% (16 out of 22 papers) of the selected publications were released between 2021 and 2025. This high density reflects accelerating industrial pressure to formalize lifecycle standards as real-world enterprise deployments scale [3,6,11]. While the initial era (2015–2018) focused heavily on highlighting risks and quantifying system debt (e.g., [5]), the modern era (2021–2025) shifts entirely toward automated orchestration paradigms, sustainable Green MLOps [12], and advanced Large Language Model engineering [13]. As illustrated in Figure 1, the temporal distribution reveals a clear inflection point in 2020–2021, corresponding to the widespread industrialization of MLOps tooling.

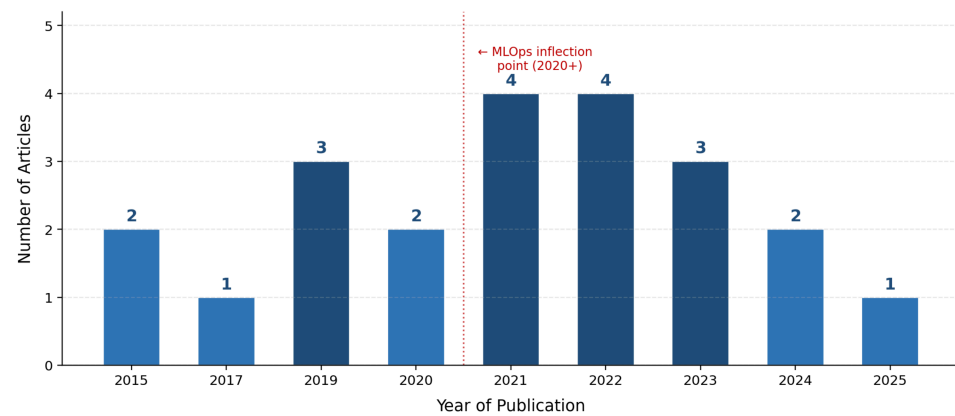


Figure 1. Temporal distribution of ML lifecycle publications included in this review ($n = 22$), showing an accelerating publication trend from 2020 onwards.

3.1.2. Distribution by Content Domain and Functional Scope

As illustrated in Figure 2, the total thematic coverage across the 22 selected studies isolates three dominant functional archetypes [3,4,11,14–18]:

- Automation and Infrastructure Engineering (40.9%, $n = 9$): Heavy emphasis on CI/CD orchestration, automated continuous retraining loops, CD4ML structures, and AutoML pipelines [10,15,17,19,20].
- Conceptual and Architectural Meta-Models (31.8%, $n = 7$): Theoretical mapping of macro-level operational phases, cross-functional organizational roles, and tool integration layers [3,8,21–23].
- Governance, Security, and Quality Control (27.3%, $n = 6$): Critical focus on model risk assessment, threat modeling, regulatory compliance, testing metrics, and Explainable AI components [14,16,17,24].

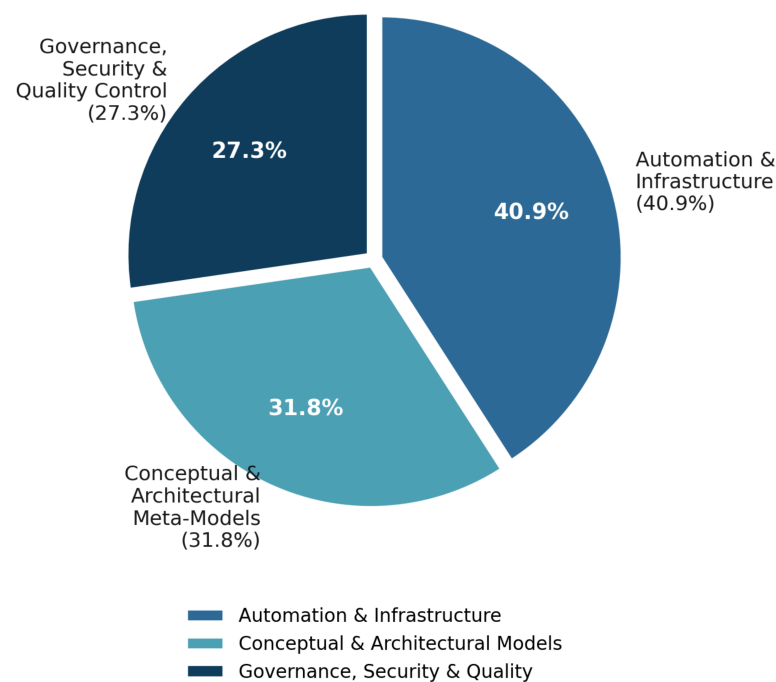


Figure 2. Distribution of the 22 included studies across three functional archetypes, illustrating the dominant focus on Automation and Infrastructure Engineering (40.9%; [10,15,17,19,20]), followed by Conceptual and Architectural Meta-Models (31.8%; [3,8,21–23]) and Governance, Security, and Quality Control (27.3%; [14,16,17,24]).

Table 1 below presents the full classification of the 22 included studies by thematic focus area, methodology, and key supporting references.

Table 1. Classification of the 22 included studies by functional focus and methodology.

Focus Area	<i>n</i> (%)	Primary Methodology	Key References
Lifecycle Frameworks and MLOps	7 (31.8%)	Framework design and prototyping	[3,15,17,20–22,25]
Operational Challenges and Technical Debt	5 (22.7%)	Case studies and empirical surveys	[2,4,5,9,11]
Security, Ethics and Governance	4 (18.2%)	Risk assessment and analytical modeling	[14,16,17,24]
Domain-Specific Applications	4 (18.2%)	Experimental and technical analysis	[13,19,23,26–28]
Automation and Infrastructure/AutoML	2 (9.1%)	CI/CD pipelines and AutoML integration	[10,20]

3.2. Deep Domain Expansion: Specialized Lifecycles

3.2.1. Healthcare and Clinical Deep Learning Pipelines

Standard enterprise MLOps models assume flexible, high-frequency continuous deployment loops [3,15]. However, as demonstrated in specialized computational medicine applications, clinical deep learning pipelines face extreme constraints due to clinical validation friction, multi-task data structures, and stringent patient privacy laws [26–28]. The lifecycle for medical imaging pipelines must incorporate highly specialized data preparation and validation checkpoints [26]. For instance, in automated brain tumor diagnosis frameworks utilizing magnetic resonance imaging (MRI) data, the pipeline must seamlessly couple multi-sequence semantic segmentation masks with radiomics extractions before passing the representations to generative AI layers for clinical report synthesis [26,28]. This creates severe lifecycle dependencies, where any upstream changes in scanner hardware calibration trigger rapid downstream data drift, corrupting multi-task classification stability [4,28].

Furthermore, continuous model updating is severely constrained by medical device regulatory compliance requirements [27,28]. Unrolled deep learning frameworks applied to sparse or limited-view biological tomography data require frozen state validation to guarantee clinical safety [27], preventing standard automated continuous deployment loops without intensive human-in-the-loop clinical re-certification. These domain-specific constraints highlight a critical gap between generic MLOps reference architectures [3,6] and the regulatory and safety demands of clinical AI deployment [26,28].

3.2.2. Natural Language Processing (NLP) and Large Language Model (LLM) Pipelines

Unlike classical tabular models that suffer primarily from numerical feature drift [4,24], Large Language Model (LLM) and generative AI lifecycles are vulnerable to complex structural and semantic misalignments [13]. High-dimensional textual embeddings are highly sensitive to schema drift and subtle semantic drift, where changes in user vocabulary or underlying cultural contexts degrade prompt template effectiveness and vector database retrieval precision [13]. In specialized enterprise architectures designed to map shifting academic curricula against real-world labor market dynamics, the NLP pipeline must continually align highly fluid, non-standardized multi-lingual training texts [13].

This necessitates the introduction of a permanent, automated semantic alignment verification step within the monitoring phase of the generative lifecycle [13] to proactively detect behavioral degradation and prevent toxic hallucination cascades before the model undergoes automated fine-tuning or retrieval-augmented regeneration. These challenges extend the technical debt taxonomy documented by Sculley et al. [5] into a qualitatively new domain where non-deterministic semantic behavior cannot be monitored using classical numerical drift statistics [4,24].

3.3. Key Findings Across the Literature Cluster

- The Core Infrastructure Hypothesis [2,5]: Seminal studies prove that the actual mathematical ML algorithm code represents merely a tiny fraction of a production system [5]. The vast majority of an enterprise machine learning system consists of massive, fragile supporting infrastructure—configuration management, data collection, feature extraction, metadata tracking, and physical orchestration tooling [2,7].
- Operational Maturity Realization [3,15]: Establishing operational stability requires transitioning to formalized Continuous Delivery for Machine Learning (CD4ML) architectures [15]. These paradigms treat code, data versions, and trained model states as co-equal, version-controlled artifacts within fully automated build pipelines [3,15,25].
- Systemic Debt Mitigation [7,14]: Quantifying operational safety requires strict adherence to standardized test scores and readiness rubrics [14], ensuring that data validation, skew detection, and integration assertions run continuously rather than post-incident [7]. Communication gaps between software engineers and data scientists further compound this debt [9,11].
- Distributed Provenance and Lineage [17,18]: Managing high-volume data lineage demands robust distributed provenance frameworks [18]. Blockchain-backed validation ledgers represent a compelling architectural pattern to guarantee immutable data tracking across decentralized pipelines [17], provided that latency constraints are systematically controlled.
- Sustainability and Green MLOps [12]: Recent literature highlights the growing importance of carbon footprint tracking during continuous training cycles [12]. Environmental sustainability is emerging as a design constraint alongside accuracy and latency, driven by regulatory and ethical imperatives [3,12].

4. Discussion and Conceptual Synthesis

4.1. Comparative Analysis: CRISP-DM vs. Modern MLOps

To explicitly address the structural deficiencies in legacy paradigms, Table 2 maps out the precise operational and architectural evolutionary leap required between the historical CRISP-DM workflow and contemporary, automated production MLOps structures [1,3,6,10].

Table 2. Structural comparison: Legacy CRISP-DM vs. Modern Automated MLOps.

Dimension	Legacy CRISP-DM	Modern Automated MLOps	Key Refs.
Core Workflow Structure	Linear, cyclical; heavily manual	Continuous, real-time automated loops (CI/CD/CM); event-driven triggers	[3,15,25]
Primary Artifact Focus	Static model file; isolated offline evaluation	Unified trinity: Versioned Code + Data + Models	[3,10,15]
Deployment Strategy	Manual, monolithic code translation; team hand-offs	Automated microservices orchestration; containerized APIs	[3,6,15]
Post-Production Monitoring	Ad hoc, periodic manual data audits	Automated, continuous telemetry; real-time drift detection	[4,18,24]
Retraining Trigger	Manual when accuracy drops below the threshold	Automated pipeline triggered by drift alerts	[3,4,18]

4.2. The Adaptive Lifecycle Framework (ALF): A Four-Phase Conceptual Model

Synthesizing the core engineering requirements documented across the 22-paper corpus [1,3–5], this section introduces the Adaptive Lifecycle Framework (ALF)—a four-phase, feedback-driven conceptual model that addresses the industrial production gap [4,5] by explicitly binding technical pipeline automation with continuous systemic governance

controls [3,12,16]. Unlike the static, linear structure of CRISP-DM [1,10], the ALF is designed as a closed-loop cycle, as illustrated in Figure 3.

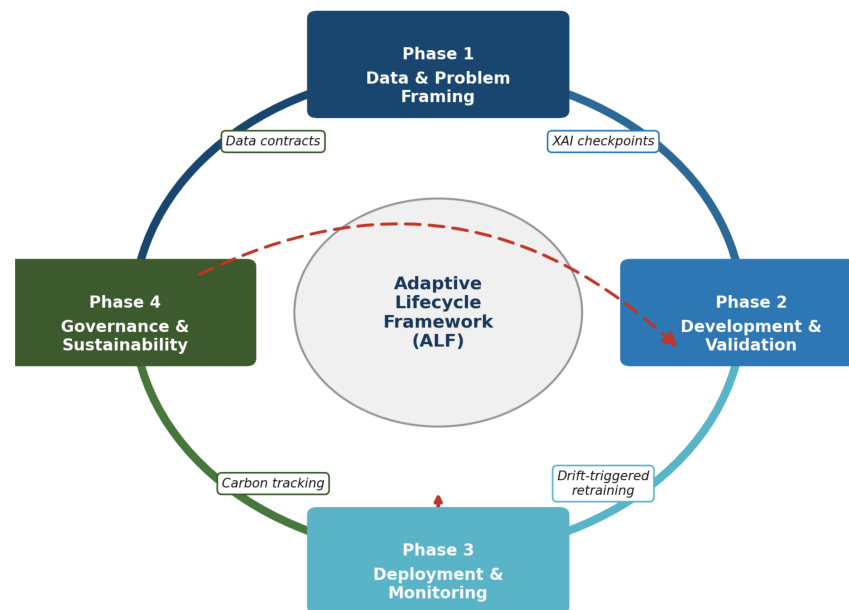


Figure 3. The Adaptive Lifecycle Framework (ALF): a four-phase closed-loop cycle integrating dynamic re-training, XAI governance, and sustainability monitoring—Phase 1: Data and Problem Framing [1,3,18]; Phase 2: Development and Validation [3,14,16]; Phase 3: Deployment and Monitoring [3,4,15,18]; Phase 4: Governance, Sustainability and Compliance [12,14,16,17]. Red arrows denote the two feedback-driven revalidation paths that close the loop: the drift-triggered re-training path (Phase 3 → Phase 1) and the compliance-driven revalidation path (Phase 4 → Phase 2).

The four phases of the ALF are defined as follows:

- Phase 1—Data and Problem Framing [1,3,18]: Defines the ML objective, establishes data contracts, and embeds an initial XAI checkpoint to document expected feature relevance distributions before model development begins. Data versioning and lineage tracking are activated from this phase [18].
- Phase 2—Development and Validation [3,14,16]: Iterative model development with mandatory bias audits, security reviews, and XAI profiling (e.g., SHAP/LIME value matrices [16]) prior to promotion. Readiness is assessed against standardized scoring grids [14].
- Phase 3—Deployment and Monitoring (Dynamic Adaptability) [3,4,15,18]: CI/CD-driven deployment with continuous real-time telemetry monitoring. Statistical distances (e.g., Kolmogorov–Smirnov statistics, Population Stability Indices) are computed to detect data and concept drift [4,18]. When drift thresholds are crossed, the system autonomously triggers re-training and feeds back into Phase 1, maintaining continuous improvement without manual intervention [3,15].
- Phase 4—Governance, Sustainability and Compliance [12,14,16,17]: Tracks the carbon footprint per training cycle [12], maintains an immutable blockchain-backed metadata lineage ledger [17,18], and enforces regulatory compliance (e.g., the EU AI Act). No model can re-enter Phase 1 without passing Phase 4 compliance checks, creating a closed governance loop.

Table 3 summarizes the key distinctions between the ALF and the two main reference models (CRISP-DM and the Kreuzberger et al. MLOps architecture):

Table 3. Structural comparison of the ALF against CRISP-DM and MLOps reference architectures.

Dimension	CRISP-DM	Kreuzberger MLOps [3]	ALF (This Work)
Re-training trigger	Manual	Scheduled or manual	Automated, drift-triggered
Explainability	Not embedded	Optional add-on	Phase gate at Ph.1 and Ph.2
Sustainability metric	Absent	Absent	Carbon tracking (Ph.4)
Regulatory compliance	Out of scope	Partial	Mandatory gate (Ph.4)

The ALF is presented as a conceptual synthesis framework derived from the 22-paper corpus; empirical validation through a pilot case study is identified as a primary direction for future research.

5. Conclusions

This systematic literature survey has rigorously synthesized the empirical evolution of the Machine Learning Lifecycle from 2015 to 2025 across 22 primary benchmark studies [1–12,14–25]. By analyzing modern MLOps architectures [3,6,15], hidden technical debt [5,7], and specialized constraints across computational medicine [26–28] and generative AI [13], this paper addresses the core historical limitations of non-standardized workflows. The comparative mapping against CRISP-DM [1,10] establishes a clear blueprint for the future of automated, real-time production engineering. The four-phase Adaptive Lifecycle Framework (ALF) introduced in Section 4.2 demonstrates that resolving the industrial production gap [4,5] requires the uncompromised unification of event-driven CI/CD pipelines [3,15,25], integrated Explainable AI governance [16,24], and standardized data lineage tracking [17,18]. Future operational research must focus on empirically validating the ALF and optimizing governance wrappers to minimize computational overhead while guaranteeing robust adherence to emerging global AI safety standards [12].

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